

## FURTHER MECHANICS / CENTRE OF MASS

# Centre of Mass of Systems of Particles

Questions by topic

9 questions · 57 marks

**Instructions**

Unless a question states otherwise, take  $g = 9.8 \text{ m s}^{-2}$  wherever a numerical value of  $g$  is required. Solutions also give final answers for  $g = 10 \text{ m s}^{-2}$  where these differ.

## Questions

|            |          |    |
|------------|----------|----|
| Question 1 | 6 marks  | 2  |
| Question 2 | 11 marks | 3  |
| Question 3 | 7 marks  | 4  |
| Question 4 | 5 marks  | 5  |
| Question 5 | 4 marks  | 6  |
| Question 6 | 7 marks  | 7  |
| Question 7 | 6 marks  | 8  |
| Question 8 | 5 marks  | 9  |
| Question 9 | 6 marks  | 10 |

1. Three particles of masses  $m$ ,  $2m$  and  $km$  are placed at the points  $A(a, 0)$ ,  $B(-2a, 3a)$  and  $C(2a, -a)$  respectively, where  $k > 0$ .

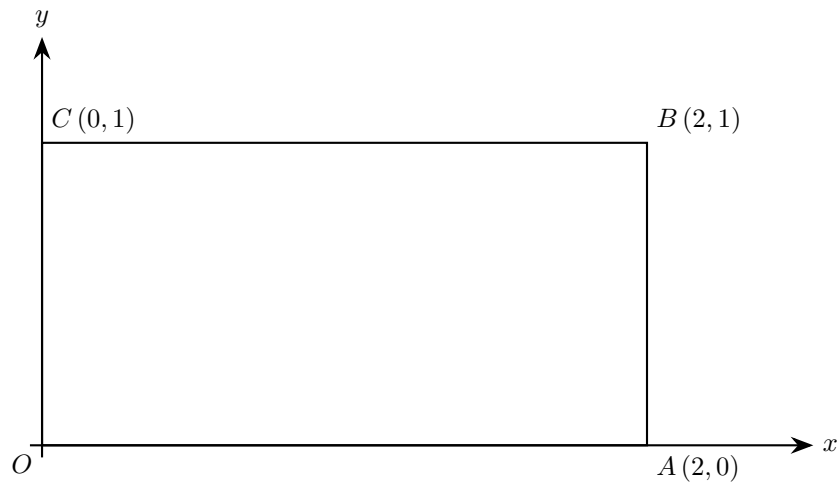
The centre of mass of the three particles is the point with coordinates  $(\bar{x}, \bar{y})$ .

- (a) (i) Find  $\bar{x}$  in terms of  $a$  and  $k$ .  
(ii) Find  $\bar{y}$  in terms of  $a$  and  $k$ .

[4]

- (b) Given that the centre of mass lies on the circle with centre  $(a, 0)$  and radius  $\frac{\sqrt{2}}{2}a$ , find the possible values of  $k$ .

[2]



2. The diagram shows a light rectangular lamina  $OABC$  where  $O$  is the origin of the coordinate system in which the points  $A$ ,  $B$  and  $C$  have coordinates  $(2, 0)$ ,  $(2, 1)$  and  $(0, 1)$  respectively. Coordinates refer to the axes shown and the units are metres.

The following three forces are applied to the lamina.

$$\begin{pmatrix} p \\ 2q \end{pmatrix} \text{ N at the point } (1, 0)$$

$$\begin{pmatrix} -q \\ p \end{pmatrix} \text{ N at the point } C$$

$$\begin{pmatrix} -3 \\ -6 \end{pmatrix} \text{ N at the point } (t, 1), \text{ where } 0 \leq t \leq 2$$

The three forces form a couple of magnitude  $3 \text{ N m}$  in the anticlockwise direction.

- (a) Determine the values of  $p$ ,  $q$  and  $t$ .

[5]

The three forces are removed. The lamina is in the same position as shown in the diagram.

Three particles are now attached to certain points on the lamina as follows.

- (b) Find the coordinates of the composite body's centre of mass. Give your answer in terms of  $\lambda$ , where the particles are
- a particle of mass  $2 \text{ kg}$  at the point  $(2\lambda^2, \lambda)$ , where  $0 \leq \lambda \leq 1$
  - a particle of mass  $1 \text{ kg}$  at the point  $A$
  - a particle of mass  $1 \text{ kg}$  at the point  $C$

[3]

- (c) Show that, as  $\lambda$  varies, the centre of mass of the combined object lies on the curve

$$x = 4y^2 - 2y + k$$

where  $k$  is a constant to be determined.

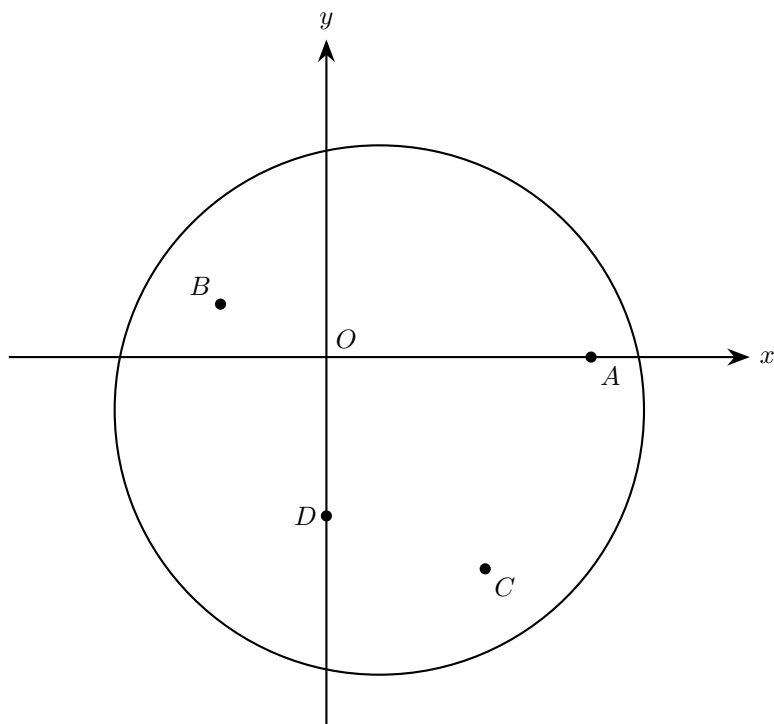
[3]

- 3.** Three particles of masses  $2m$ ,  $5m$  and  $3m$  are placed at the points  $(1, -4)$ ,  $(-3, 2)$  and  $(q, 2q - 1)$  respectively.

The value of  $q$  is such that the distance of the centre of mass of the three particles from the point  $(1, 0)$  is as small as possible.

Find the value of  $q$ .

[7]



4. Four sensors are attached to a circular plate.

The plate is to be modelled as a uniform lamina and the four sensors as four particles.

The diagram shows the lamina, the four particles  $A$ ,  $B$ ,  $C$  and  $D$ , and the coordinate axes.

The lamina has mass 6 kg and its centre of mass is at the point  $(1, -1)$ .

Particle  $A$  has mass 4 kg and is at the point  $(5, 0)$ .

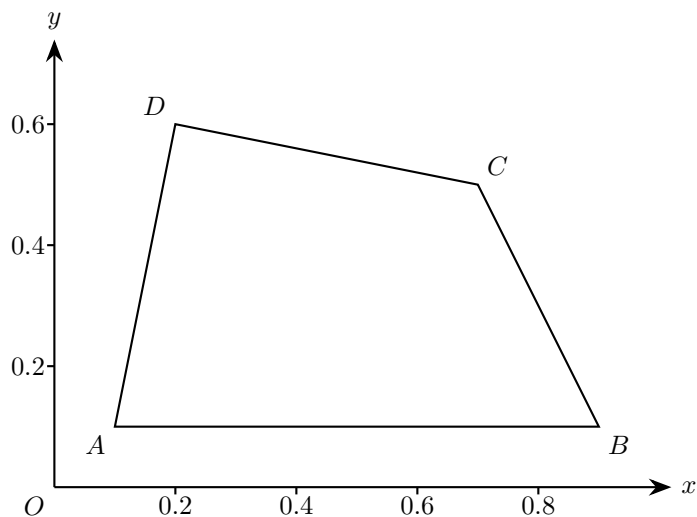
Particle  $B$  has mass 3 kg and is at the point  $(-2, 1)$ .

Particle  $C$  has mass 5 kg and is at the point  $(3, -4)$ .

Particle  $D$  has mass 2 kg and is at the point  $(0, -3)$ .

Find the coordinates of the centre of mass of the system of plate and sensors.

[5]



5. Particles  $A$ ,  $B$ ,  $C$  and  $D$  of masses 1 kg, 2 kg, 3 kg and 4 kg respectively are joined by light rigid rods to form the quadrilateral frame shown in the diagram. The particles are at the points  $A(0.1, 0.1)$ ,  $B(0.9, 0.1)$ ,  $C(0.7, 0.5)$  and  $D(0.2, 0.6)$ , where coordinates are measured in metres.

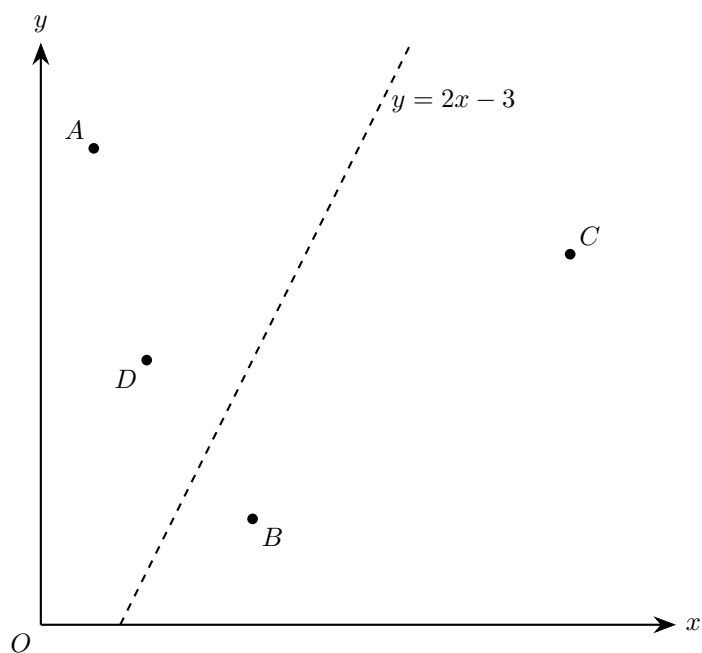
$G$ , the centre of mass of the frame, has coordinates  $(\bar{x}, \bar{y})$ .

- (a) Show that  $\bar{y} = 0.42$ . Find  $\bar{x}$ . [3]
- (b) Using your answer to part (a), explain why the frame cannot be in equilibrium in a horizontal plane if it is supported by a single peg touching the frame at one point. [1]

6. Four particles of mass  $m$ ,  $2m$ ,  $4m$  and  $\lambda m$  are placed at points with coordinates  $(0, 0)$ ,  $(2, 3)$ ,  $(1, -1)$  and  $(4, 2)$  respectively. The centre of mass of the system is at  $(2, k)$ .

(a) Show that  $\lambda = 3$ . [4]

(b) Calculate the value of  $k$ . [3]



7. The diagram shows four particles,  $A$ ,  $B$ ,  $C$  and  $D$ , fixed in a horizontal plane which contains the  $x$ - and  $y$ -axes.

Particle  $A$  has mass 3 kg and is attached at the point  $(1, 9)$ .

Particle  $B$  has mass 4 kg and is attached at the point  $(4, 2)$ .

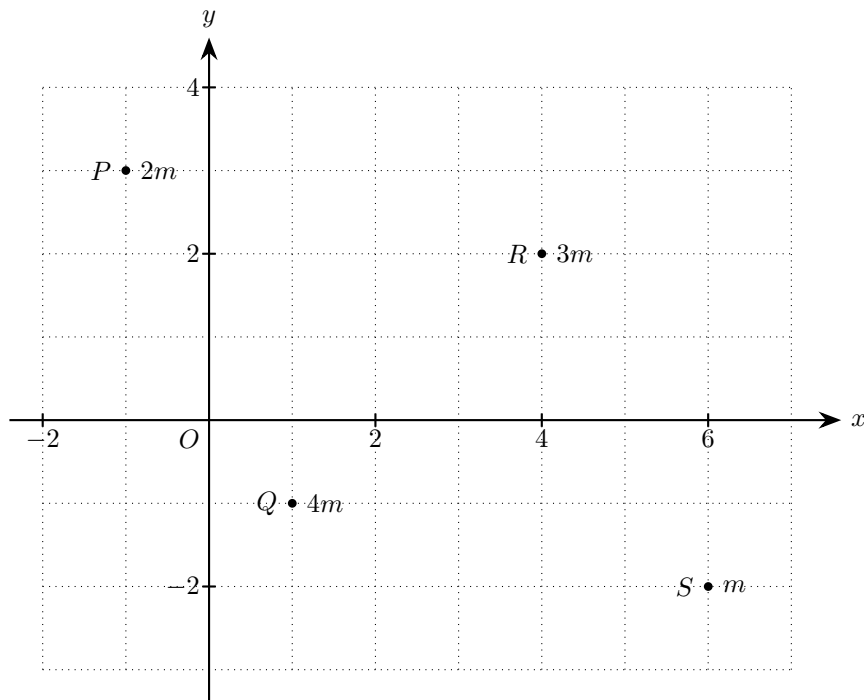
Particle  $C$  has mass 5 kg and is attached at the point  $(10, 7)$ .

Particle  $D$  has mass  $m$  kg and is attached at the point  $(2, 5)$ .

The centre of mass of the four particles lies on the line  $y = 2x - 3$ .

Find the value of  $m$  and the coordinates of the centre of mass.

[6]



8. The diagram shows a system of four particles of masses  $2m$ ,  $4m$ ,  $3m$  and  $m$  situated in the  $x$ - $y$  plane at the points  $P(-1, 3)$ ,  $Q(1, -1)$ ,  $R(4, 2)$  and  $S(6, -2)$  respectively.

Find the coordinates of the centre of mass of the system.

[5]

9. Four particles of masses  $m$ ,  $2m$ ,  $3m$  and  $km$  are positioned at the points with coordinates  $(-a, a)$ ,  $(2a, -2a)$ ,  $(0, 0)$  and  $(\mu a, 2\mu a)$  respectively, where  $k$  and  $\mu$  are constants.

The centre of mass of the four particles is at the point with coordinates  $(2a, 3a)$ .

(a) Find the value of  $k$ . [3]

(b) Find the value of  $\mu$ . [3]